YEAR 10 MATHS · CHAPTER 8 · MIND MAP

Probability and counting

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1 Review of probability

$P(A) = \dfrac{\text{favourable}}{\text{total}}$ and $P(A') = 1 - P(A)$.

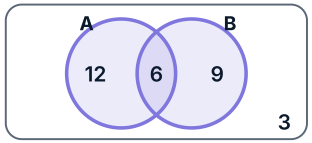
52 cards: $P(\text{heart}) = \frac{13}{52} = \frac{1}{4}$

$P(\text{not an ace}) = \frac{48}{52} = \frac{12}{13}$

2 Venn diagrams and two-way tables

$A \cap B$: both. $A \cup B$: either. A': not A.

30 people: $n(A \cup B) = 27$, $P(A') = \frac{12}{30} = \frac{2}{5}$



3 The addition rule

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$0.5 + 0.4 - 0.2 = 0.7$

Mutually exclusive: $P(A \cap B) = 0$

4 Conditional probability

$P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$: only look inside B.

From the Venn in card 2: $P(A \mid B) = \frac{6}{15} = \frac{2}{5}$

5 Arrays for two steps

List every outcome in a table.

Two dice: 36 outcomes.

$P(\text{double}) = \frac{6}{36} = \frac{1}{6}$

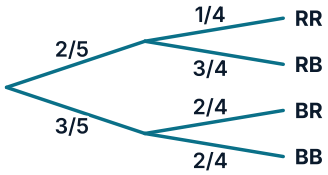
$P(\text{sum} \geq 10) = \frac{6}{36} = \frac{1}{6}$

6 Tree diagrams

Multiply along the branches, add the branches you want.

2 red, 3 blue, take 2 without replacement:

$P(\text{one of each}) = \frac{2}{5} \cdot \frac{3}{4} + \frac{3}{5} \cdot \frac{2}{4} = \frac{3}{5}$



7 Independent events

Independent if $P(A \cap B) = P(A) \times P(B)$, so $P(A \mid B) = P(A)$.

Coin and die: $P(\text{H and } 6) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

8 Counting and factorials

Multiply the number of choices at each step.

$n! = n \times (n-1) \times \cdots \times 1$

3 shirts and 4 pairs of pants: 3 × 4 = **12** outfits

$5! = 120 \quad 0! = 1$

9 Arrangements

Order matters: ${}^nP_r = \dfrac{n!}{(n-r)!}$

Arrange 3 of 5 books: $5 \times 4 \times 3 = 60$

Arrange the letters of MATHS: $5! = 120$

10 Selections

Order doesn't matter: ${}^nC_r = \dfrac{n!}{r!(n-r)!}$

Choose 3 from 5: ${}^5C_3 = 10$

A pair from 6 people: ${}^6C_2 = 15$

Watch out: a team is a selection; a 1st, 2nd and 3rd place is an arrangement.

11 Counting in probability

$P = \dfrac{\text{favourable selections}}{\text{total selections}}$

Pick 3 from 5 boys and 4 girls: $P(\text{all girls}) = \dfrac{{}^4C_3}{{}^9C_3} = \dfrac{4}{84} = \dfrac{1}{21}$